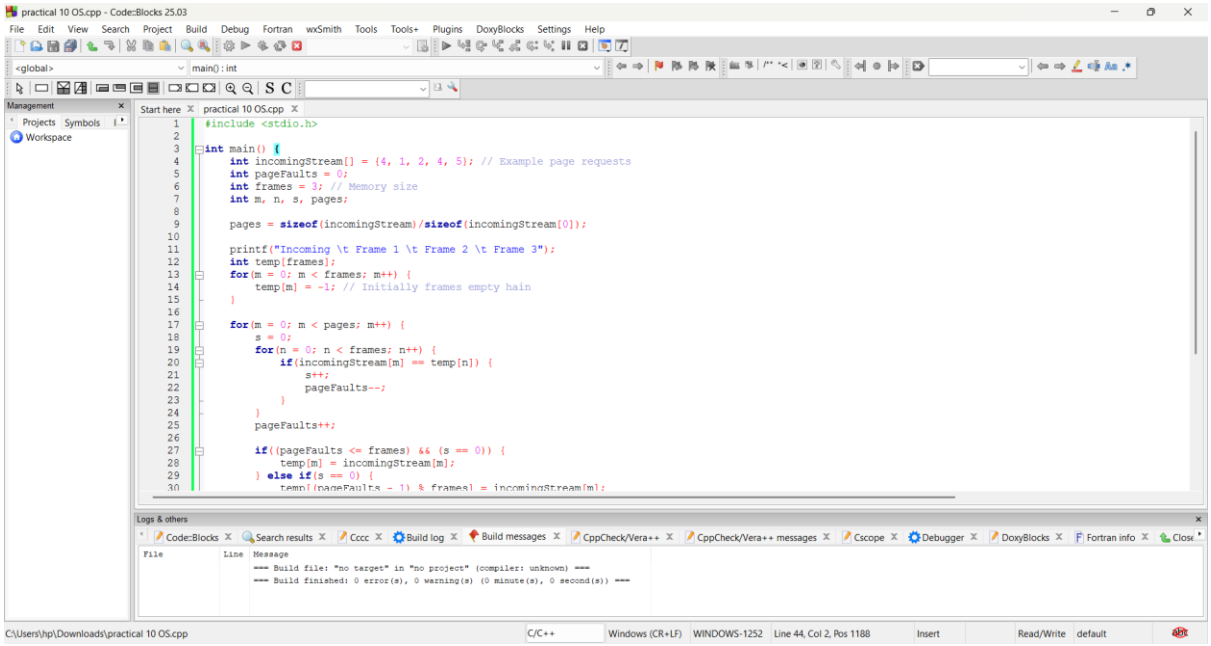
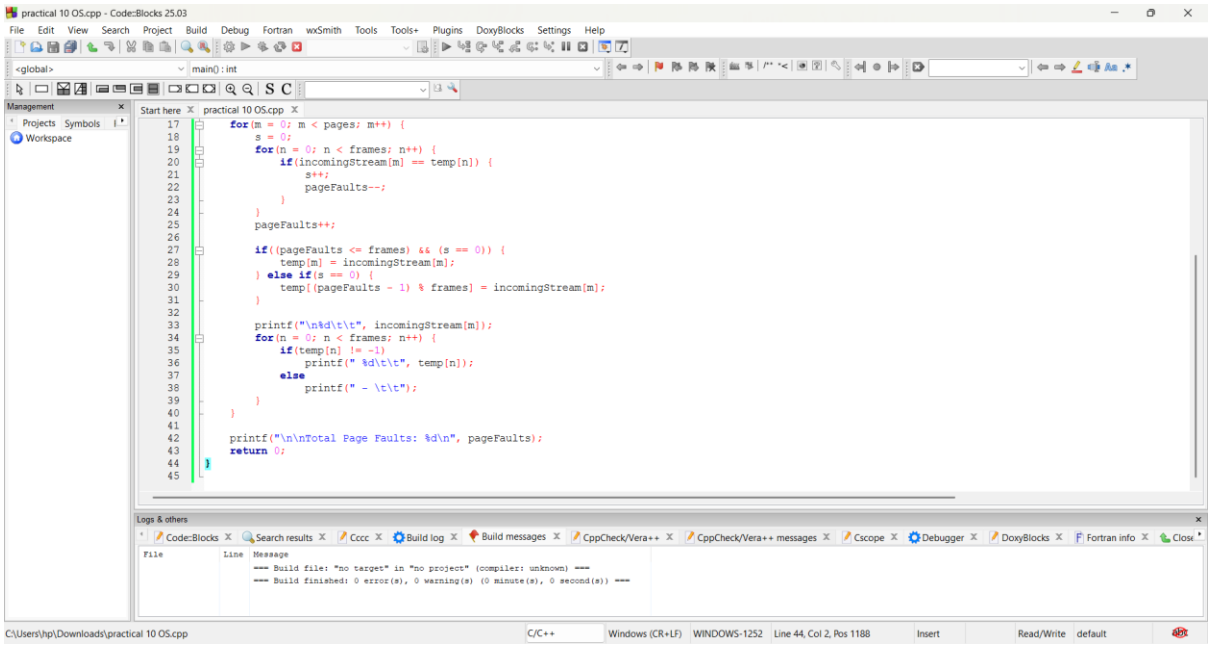
Aim: Implementing a Virtual Memory System with Demand Paging where pages are loaded into memory only when needed, optimizing memory usage and minimizing page faults.



```
#include <stdio.h>

int main() {
    int incomingStream[] = {4, 1, 2, 4, 5}; // Example page requests
    int pageFaults = 0;
    int frames = 3; // Memory size
    int m, n, s, pages;

    pages = sizeof(incomingStream)/sizeof(incomingStream[0]);

    printf("Incoming \t Frame 1 \t Frame 2 \t Frame 3");
    int temp[frames];
    for(m = 0; m < frames; m++) {
        temp[m] = -1; // Initially frames empty hain
    }

    for(m = 0; m < pages; m++) {
        s = 0;
        for(n = 0; n < frames; n++) {
            if(incomingStream[m] == temp[n]) {
                s++;
                pageFaults--;
            }
        }
        pageFaults++;

        if((pageFaults <= frames) && (s == 0)) {
            temp[m] = incomingStream[m];
        } else if(s == 0) {
            temp[(pageFaults - 1) % frames] = incomingStream[m];
```



```
    for(m = 0; m < pages; m++) {
        s = 0;
        for(n = 0; n < frames; n++) {
            if(incomingStream[m] == temp[n]) {
                s++;
                pageFaults--;
            }
        }
        pageFaults++;

        if((pageFaults <= frames) && (s == 0)) {
            temp[m] = incomingStream[m];
        } else if(s == 0) {
            temp[(pageFaults - 1) % frames] = incomingStream[m];
        }

        printf("\n%d\t\t", incomingStream[m]);
        for(n = 0; n < frames; n++) {
            if(temp[n] != -1)
                printf(" %d\t\t", temp[n]);
            else
                printf(" - \t\t");
        }
    }

    printf("\n\nTotal Page Faults: %d\n", pageFaults);
    return 0;
}
```

```
"C:\Users\hp\Downloads\prac

Incoming         Frame 1         Frame 2         Frame 3
4                4               -               -
1                4               1               -
2                4               1               2
4                4               1               2
5                5               1               2

Total Page Faults: 4

Process returned 0 (0x0)   execution time : 0.078 s
Press any key to continue.
```